

Milestone 2

JAVA Lab Resource Utilization Platform

Milestone Title: Utilization Monitoring & Inter-Institution Resource Sharing

Objective

The objective of Milestone 2 is to enhance the existing resource management system by implementing real-time equipment utilization monitoring, utilization visualization, inter-institution resource sharing, external booking, waitlist management, and booking optimization.

Scope

- Real-time equipment utilization monitoring
- Equipment usage visualization
- Heatmap generation
- Idle equipment detection
- Inter-institution equipment sharing
- External booking system
- Demand analysis
- Waitlist management
- Booking optimization

Functional Requirements

1. Real-Time Equipment Utilization Monitoring: Track equipment status (Available, Booked, In Use, Under Maintenance), automatic updates, live dashboard.
2. Utilization Heatmap & Idle Time Detection: Daily/weekly/monthly heatmaps, idle equipment detection, color coding.

3. Inter-Institution Resource Sharing: Institution registration, shared equipment catalog, request approval/rejection workflow.
4. External Booking Management: External registration, booking, approval, cancellation and access management.
5. Utilization Rate & Demand Analysis: Utilization %, total bookings, peak usage and demand analysis. Formula: $\text{Utilization Rate} = (\text{Hours Used} / \text{Total Available Hours}) \times 100$.
6. Waitlist Management & Booking Optimization: Automatic waitlist, queue management, notifications, alternative slot suggestions.

Frontend Modules

- Utilization Dashboard
- Equipment Status Page
- Utilization Heatmap
- Shared Equipment Page
- External Booking Page
- Waitlist Page
- Booking Approval Page

Backend Modules

- Equipment Status Service
- Booking Service
- Utilization Service
- Sharing Service
- Waitlist Service
- Notification Service
- External Booking Service

Database Tables

- Equipment
- Booking
- Booking History

- Utilization Log
- Institution
- Shared Equipment
- Sharing Request
- External User
- Waitlist
- Notification

Technologies

- Frontend: React.js, HTML, CSS, JavaScript
- Backend: Spring Boot, Java, REST APIs
- Database: MySQL
- Tools: GitHub, VS Code, Postman, Figma

Expected Deliverables

Real-time utilization monitoring dashboard, utilization heatmap, idle equipment detection, inter-institution resource sharing, external booking and access management, waitlist management, booking optimization, and demand analysis.

Learning Outcomes

Develop monitoring systems, utilization analysis, sharing workflows, booking optimization, demand analysis, dashboard visualization, frontend-backend integration.

Conclusion

Milestone 2 enhances the platform with real-time utilization monitoring, heatmap visualization, idle equipment detection, inter-institution resource sharing, external booking, waitlist management, demand analysis, and booking optimization to improve laboratory resource utilization.